

IMAGE PROCESSING PROJECT

---

# Bindu:

## an image workspace

---

Channel analysis, reconstruction  
and image processing

Python / Streamlit / NumPy / OpenCV

# Feature overview

---

- |           |                            |                                          |
|-----------|----------------------------|------------------------------------------|
| <b>01</b> | <b>Channel analyzer</b>    | RGB previews, histograms and FFT spectra |
| <b>02</b> | <b>Reconstruction</b>      | Recover selected color channels          |
| <b>03</b> | <b>Color spaces</b>        | Compare RGB and YCbCr                    |
| <b>04</b> | <b>Fourier compression</b> | Prune coefficients and decode archives   |
| <b>05</b> | <b>Image filters</b>       | Smooth or sharpen image details          |

Upload an image or choose a sample, then explore the processing results.

# Channel analyzer

---

Inspect each RGB component in the spatial and frequency domains

- **Channel previews:** isolate red, green and blue contributions.
- **Histograms:** compare pixel intensity distributions.
- **2-D FFT spectra:** inspect spatial frequency content.

$$F_c(u, v) = \mathcal{F}\{I_c(x, y)\}, \quad c \in \{R, G, B\}$$

Log-magnitude display makes weaker frequency components easier to see.

# Partial image reconstruction

---

## Keep one channel or a pair

- Choose R, G, B, R+G, R+B or G+B
- Apply the inverse FFT to selected channels
- Set omitted channels to zero

## Full complex spectra

$$\hat{I}_c = \begin{cases} \mathcal{F}^{-1}(F_c), & c \text{ selected,} \\ 0, & c \text{ omitted.} \end{cases}$$

Example: R+G preserves red and green values and removes blue.

Selected channels retain both magnitude and phase.

# Color spaces: RGB and YCbCr

---

## RGB

Red, green and blue intensities describe each pixel.

## YCbCr

Y represents luma. Cb and Cr represent color differences.

Shared plotting scales support comparison. Mean removal is on by default for spectra. This tool does not compress or subsample the image.

## Full-range BT.601 conversion

$$Y = 0.299R + 0.587G + 0.114B$$

Neutral chroma:  $Cb = Cr = 128$ .

Compare grayscale previews, histograms and FFT spectra.

# Fourier compression

1. Compute a 2-D FFT per channel
2. Rank conjugate pairs by energy
3. Keep DC and the strongest pairs
4. Store indices and coefficients

$$\frac{\sum_{k \in K} |F(k)|^2}{\sum_k |F(k)|^2} \geq \frac{p}{100}$$

Decode with an inverse FFT.

## Measured quality–size tradeoff

| Energy | Archive   | PSNR     |
|--------|-----------|----------|
| 95%    | 2.7 KiB   | 17.09 dB |
| 99%    | 15.0 KiB  | 24.09 dB |
| 99.9%  | 193.8 KiB | 34.20 dB |

Higher PSNR means less pixel error.  
Energy retention is not a size ratio.

Measured with the project codec on inputs/test.jpg resized to 256 × 366 RGB pixels. Raw size: 274.5 KiB. One example, not a general benchmark.

# Image filters

---

## Smoothing

- Gaussian blur: weighted neighbors
- Box blur: local averaging

Larger blur settings smooth more detail.

Compare R-only, G-only, B-only and all-channel filtering.

Strong sharpening can amplify noise and produce halos.

## Sharpening

- Laplacian: emphasize local changes
- Unsharp mask: add detail back

$$I_{\text{sharp}} = I + a(I - G_{\sigma} * I)$$

$G_{\sigma} * I$ : Gaussian-blurred image.

$a$ : sharpening amount.

# Thank you

Questions and discussion

---

Exploring how channel selection, compression  
and filtering affect images.